Project 2
Pizza Sales Analysis Report

INGRADE

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[Link to github](#)

[Link to dataset](#)

Overview

The fast-paced food delivery industry relies heavily on efficiency and customer satisfaction.

This analysis of pizza sales data from multiple restaurants aims to explore key delivery performance metrics, popular choices, and operational bottlenecks.

With rising consumer expectations and increased traffic challenges, it's crucial to assess how various factors — like traffic, time of day, and pizza preferences — influence delivery performance and customer behavior.

Objective

This project analyzes delivery and order data for a pizza business to:

- Identify peak sales periods and best-performing pizza types.
- Evaluate delivery performance and delays.
- Understand how external factors like traffic and peak hours affect operations.
- Discover customer preferences (size, toppings, payment).
- Recommend actions to optimize delivery efficiency and improve customer satisfaction.

Why This Project?

Pizza delivery is a competitive market where speed, product variety, and customer experience matter.

Delays can damage brand perception, and understanding ordering behavior helps target promotions and operational improvements.

This project helps uncover performance bottlenecks and customer trends to inform better business decisions.

Tools Used

- MySQL Workbench

About the Dataset

The dataset contains pizza order records with the following relevant fields:

Column	Description
Order ID	Unique order identifier
Restaurant Name	Store where order was placed
Location	Delivery area
Order Time, Delivery Time	Time of order and actual delivery
Delivery Duration (min)	Time taken for delivery
Delay (min)	Delay in delivery
Pizza Type, Pizza Size	Product ordered
Toppings Count	Number of toppings added
Distance (km)	Distance to customer

Column	Description
Traffic Level	Traffic status (Low/Medium/High)
Is Peak Hour, Is Weekend	Boolean indicators
Payment Method	Payment method used
Estimated Duration (min)	Expected delivery duration
Traffic Impact	Numeric impact due to traffic
Order Month, Order Hour	Date-based features
Payment Category	Type of payment (Cash/Card)
Delivery Efficiency	Duration per km

Insights

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1	Monthly Sales	August has the highest number of orders.
2	Popular Pizzas	Non-veg pizzas dominate the orders; top 5 types are highly preferred.

3	Weekend Weekday	vs Weekdays have nearly 3× more orders than weekends.
4	Peak Hour Trends	Orders spike around 7 PM, indicating dinner rush.
5	Delay Patterns	All orders are marked delayed (≥1 min), suggesting unrealistic expectations or system lag.
6	Topping Preferences	Most customers prefer 3 toppings on their pizza.
7	Best Locations	Atlanta has the highest orders and good delivery time, indicating efficient service.
8	Traffic Impact	High traffic areas show the highest average delay.
9	Payment Method	Card payments are preferred over cash.
10	Pizza Size	Medium pizzas are the most frequently ordered.
11	Delivery Efficiency	Orders in high-traffic zones are significantly less efficient.
12	Worst Deliveries	Some orders show very poor delivery efficiency (long time for short distances).